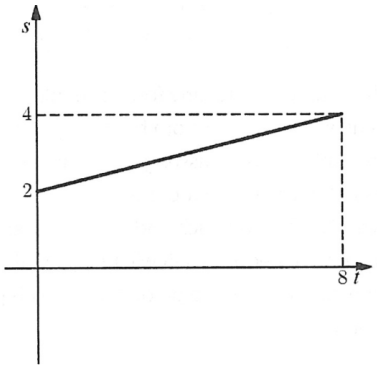
Kinematics

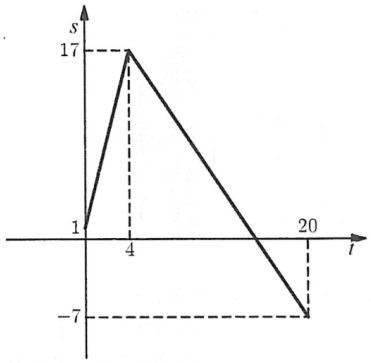
Questions 1–85

1.64 — Question 1. A battery-powered toy car is placed on a straight track. The car is switched on, so that when it contacts the track, it instantly moves with constant velocity. The following graph shows the displacement s (metres), over time t (seconds) of the toy car:



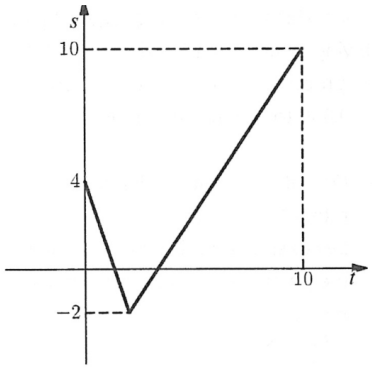
a) How far along the track is the car initially placed?
b) How far along the track is the car when 8 seconds have passed?
c) Find the velocity of the car.
d) How far along the track is the car after 5 seconds?

1.64 — Question 2. Displacement s (metres) over time t (seconds) for a particle is given in the graph below:



Find
a) the distance travelled over the first 20 seconds.
b) the displacement over the first 20 seconds.
c) the average speed of the particle.
d) the velocity for the first part of the motion.
e) the velocity for the second part of the motion.
f) the average velocity of the particle.

1.64 — Question 3. Displacement s (metres) over time t (seconds) for a particle is given in the graph below:



Find, after the first 10 seconds,

a) the distance travelled by the particle.
b) the displacement of the particle.
c) the average speed of the particle.
d) the average velocity of the particle.

1.64 — Question 4. Lisa runs in a straight line for 60 seconds at a constant speed of 3 ms^{-1}. She then pauses for 20 seconds before turning and running in the opposite direction for 45 seconds at 4 ms^{-1}. Draw a displacement-time graph to show the athlete's motion.

1.64 — Question 5. A child is running along a straight, flat path. He runs for 6 seconds at 2 ms^{-1}. He then stops and waits for 5 seconds, before running back along the path for 9 seconds, returning to his original starting point.

a) Sketch the displacement-time graph for the motion of the child.

b) Find the exact velocity for the final stage of the motion.

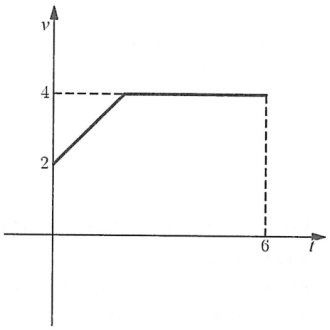
1.64 — Question 6. An object travels 20 metres at constant velocity v ms^{-1} taking t seconds. It then pauses and remains stationary for 5 seconds before returning to its original position in $\frac{1}{2}t$ seconds.

a) Sketch a displacement-time graph for the motion of the object.

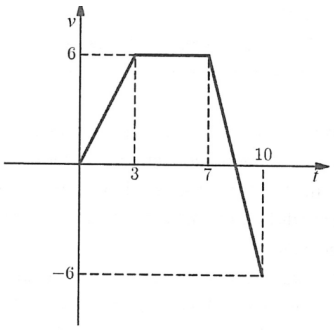
b) Given the total time the object is in motion is 6 seconds, find the value of t

1.64 — Question 7. Describe what is happening to the objects in each of the following velocity-time graphs for velocity v, in metres per second and time t, in seconds. For each stage of motion comment on the speed and where possible its position:

a)

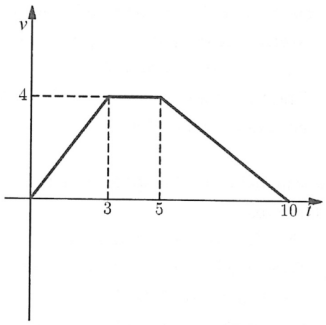


b)

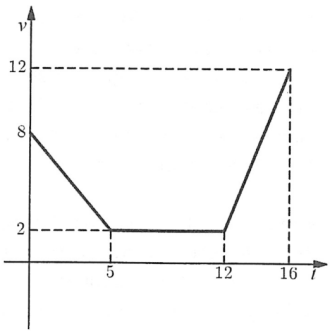


1.64 — Question 8. For each of the following velocity-time graphs below, sketch the corresponding acceleration-time graph:

a)

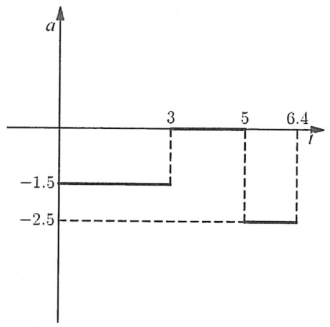


b)

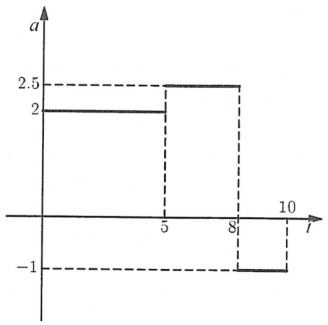


1.64 — Question 9. For each of the following acceleration-time graphs below, sketch the corresponding velocity-time graph given that the object is initially at rest:

a)



b)

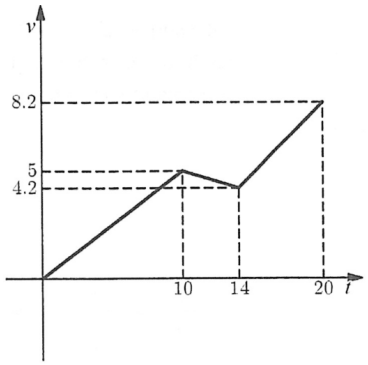


1.64 — Question 10. A car which is initially at rest, accelerates at a constant rate of 4.5 ms^{-2} for 3 seconds and then remains at this constant speed.

a) Sketch a velocity-time graph to show the first 10 seconds of the motion of the car.

b) Find the distance travelled by the car in the first 10 seconds of motion.

1.64 — Question 11. A cyclist is timed along a section of track, with velocity v recorded in metres per second at $t = 10$, $t = 14$ and $t = 20$ seconds.
The velocity-time graph for the cyclist is shown below.



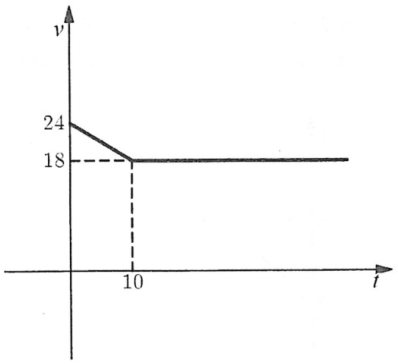
Find the

a) acceleration of the cyclist at each stage.
b) total distance travelled by the cyclist.
c) time when the cyclist is travelling at exactly 4 ms^{-1}.
d) speed the cyclist is travelling at 17 seconds.

1.64 — Question 12. Sketch a velocity-time graph which represents a car initially at rest, with:

a) constant acceleration for the first 6 seconds before reaching a fixed velocity of 10 ms^{-1} which it remains at for a further T seconds.
b) increasing acceleration for the first 2 seconds, constant acceleration for the next 3 seconds, then decreasing acceleration for the next 1 second, before reaching a fixed velocity of 10 ms^{-1} which it remains at for a further T seconds.

1.64 — Question 13. The velocity v ms^{-1} at a given time t seconds of a van in motion on a straight flat road is modelled using the velocity-time graph below:



a) Find the deceleration of the van in the first 10 seconds.
b) At 10 seconds, the van continues to travel at a constant speed of 18 ms^{-1} for a distance of 1026 m. Find the time it spends at this speed.
c) Find the total distance travelled by the van at the time given in your part b).

1.64 — Question 14. An object is moving in a straight line. It starts at 4 ms^{-1}, before accelerating uniformly to 10 ms^{-1} in 8 seconds, moving at 10 ms^{-1} for 5 seconds before finally decelerating to rest in 7 seconds.
a) Sketch a velocity-time graph which represents the motion of the object.
b) Find the distance travelled by the object.
c) Find the average velocity of the object.

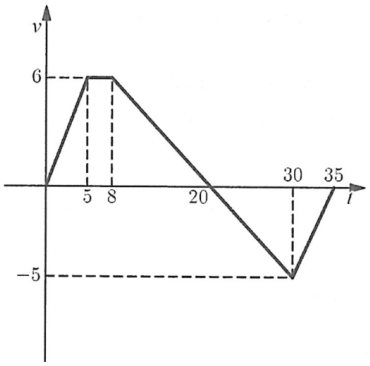
1.64 — Question 15. A van is driving along a straight stretch of motorway. It starts at 33 ms^{-1}, before decelerating to 22 ms^{-1} uniformly over 1.1 km.

a) Sketch a velocity-time graph which represents the motion of the van.
b) Find the time the van spends decelerating according to this model.

The van then maintains this speed for a further 400 metres.

c) Find the average speed of the van.
d) Two speed cameras, one at the start and one at the end of this 1.5 km stretch of road, measure the speed of the van and finds its average. Find this speed.

1.64 — Question 16. A vehicle is being driven along a flat, straight road. Its motion is modelled using the velocity-time graph below where velocity v is given in metres per second and time t is given in seconds.

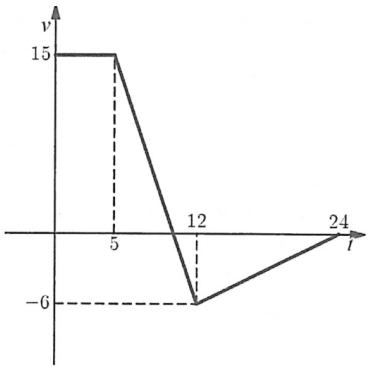


a) Find the acceleration at each stage of motion.
b) Find the total distance driven by the vehicle in the first 20 seconds.
c) Find the total distance driven by the vehicle in the final 15 seconds.
d) State the displacement of the vehicle at $t = 20$ seconds.
e) State the displacement of the vehicle at $t = 35$ seconds.
f) State the total distance driven by the vehicle.
g) Find the average speed of the vehicle over the 35 seconds.

1.64 — Question 17. A cyclist starts from rest and is cycling along a flat straight road. He accelerates at a constant acceleration of 0.5 ms^{-2} for the first 5 seconds, before accelerating at a constant acceleration of 0.3 ms^{-2} for the next 5 seconds. He then remains at a constant speed for a further 20 m.

a) Sketch a velocity-time graph to model the motion of the bicycle.
b) State the velocity of the cyclist in the third stage of the motion.
c) Find how far the cyclist has cycled since the initial start point.

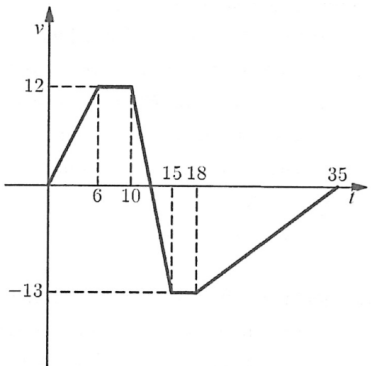
1.64 — Question 18. The velocity v ms^{-1} at a given time t seconds of a car in motion is modelled using the velocity-time graph below.



Find

a) the acceleration for all three stages of the motion.
b) the exact times at which the car, according to the graph has instantaneously stopped.
c) the total distance travelled by the car.
d) the displacement of the car from its initial position.

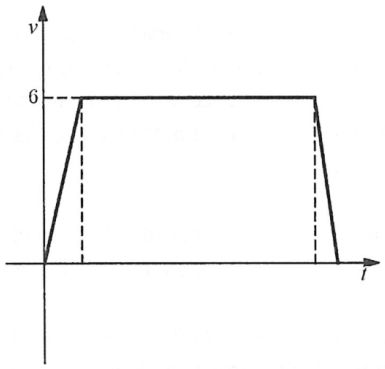
1.64 — Question 19. A ride at a theme park with a 35 second duration has a carriage that travels along a flat, straight track.
The velocity-time graph below shows the motion of the carriage in time t in seconds.



a) Find the maximum magnitude of acceleration over the 35 second ride.
b) Find the times at which the carriage is instantaneously stationary.
c) If the passengers disembark at 35 seconds, where are they in relation to the start point?

1.64 — Question 20. A bus travels from rest from a bus stop, on a straight, flat road. It drives with acceleration 0.9 ms^{-2} for 15 seconds. It is then driven at constant velocity before decelerating uniformly at 1 ms^{-2} where it comes to rest at a second bus stop which is 918 metres from the first bus stop.
a) Find the velocity of the bus when it is moving with constant velocity.
b) Sketch a velocity-time graph to represent the motion of the bus.
c) Find the time taken to get between the bus stops.

1.64 — Question 21. The velocity-time graph shows the motion of a runner running along a straight flat track of length 200 metres before coming to a stop.



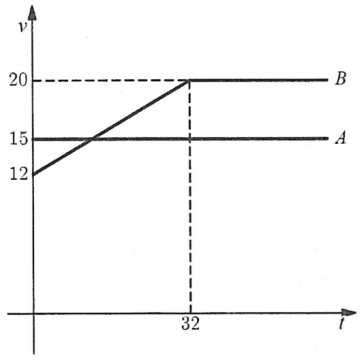
The runner accelerates uniformly from rest over the first 15 metres until reaching a speed of 6 ms^{-1} before continuing to run at 6 ms^{-1} until they complete the 200 m. A further 3 seconds later they come to rest. Find

a) the exact time the runner completes the 200 metres.
b) the total distance they run.

1.64 — Question 22. An object starts from rest and is travelling in a straight line with uniform acceleration for T seconds, travelling 24 metres. It then moves with constant velocity for 9 seconds before coming to rest uniformly in a further $\frac{3}{5}T$ seconds.

a) Sketch a velocity-time graph for the motion of the object.
b) Given the displacement of the object is 146.4 metres, find the value of T

1.64 — Question 23. Two cars, A and B are driving along a straight, flat road.
Car A is driving with constant velocity 15 ms^{-1}. Car B, which is initially driving at 12 ms^{-1}, is driving with constant acceleration until 32 seconds have passed at which it then drives at 20 ms^{-1}.
The velocity-time graph demonstrating the motion of the two cars is shown below:

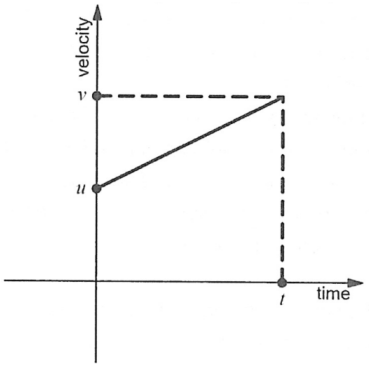


a) Find the acceleration of car B for the first 32 seconds of the motion.
b) Find the equation that gives the velocity of car B for the first 32 seconds of motion.
c) Find the exact time at which car B overtakes car A according to the model.
d) Find the distance the cars have travelled when car B overtakes car A.
e) State what the point of intersection of the two lines represents.

1.64 — Question 24. Two cars, A and B are driving along a straight, flat road. Car A is driving with constant velocity u ms^{-1}. Car B, which is initially driving at $(u-5)$ ms^{-1}, is driving with constant acceleration until T seconds have passed at which it then drives at a constant velocity. Car B overtakes car A at exactly T seconds.

a) Sketch the velocity-time graph for the two cars.
b) State the velocity of car B when it overtakes car A.
c) Given that the cars are going the same speed exactly when $t = 20$ and car B overtakes car A after driving 1000 m, find the values of T and u

1.65 — Question 1. Using the following velocity-time graph



show that

a) $v = u + at$
b) $s = \frac{1}{2}(u + v)t$
c) $s = ut + \frac{1}{2}at^2$
d) $s = vt - \frac{1}{2}at^2$

1.65 — Question 2. By substituting $v = u + at$ into $s = vt - \frac{1}{2}at^2$ show how $v^2 = u^2 + 2as$ may be derived.

1.65 — Question 3. In each case, use the constant acceleration formulae with the given values:

a) $a = 2, u = 4, v = 6$, to find t
b) $a = -2, t = 3, s = 5$, to find u
c) $s = 12, t = 4, v = 5$, to find a
d) $u = 2, a = 1.5, v = 3$, to find s
e) $s = 20, t = 8, u = 5$, to find v

1.65 — Question 4. If $s = 20, t = 6, v = 2.5$, using the constant acceleration formulae, find u and a

1.65 — Question 5. If $s = 15, v = 4.5, u = 1.5$, using the constant acceleration formulae, find a and t

1.65 — Question 6. A particle is moving in a straight line from A to B with constant acceleration 2 ms^{-2}. Its speed at A is 6 ms^{-1} and it takes 7 seconds to travel between A and B. Find

a) the speed of the particle at B

b) the distance from A to B

1.65 — Question 7. Katie is walking along a straight path of length 100 m initially at 1.1 ms^{-1} and has constant acceleration 0.05 ms^{-2}. Find the time it takes for Katie to travel the path.

1.65 — Question 8. Fasil is running along the road and for this part he can be modelled as running with constant acceleration of 0.1 ms^{-2}. He is running at 5 ms^{-1} when he passes a lamp post. A further 10 seconds later he passes a litter bin. Find

a) the distance between the lamp post and the litter bin.
b) the speed at which Fasil passes the litter bin.

1.65 — Question 9. Find the acceleration and the displacement given each of the following scenarios, where the object starts from rest:

a) $v = 4$ ms^{-1}, $t = 3$ s
b) $v = 1.4$ cms^{-1}, $t = 2$ s
c) $v = 10$ ms^{-1} giving your answer in terms of t

1.65 — Question 10. A cyclist is cycling on a straight flat road. Her initial velocity is u ms^{-1}, then she accelerates at a constant rate of a ms^{-2} for 12 seconds before reaching a speed of $3u$ ms^{-1}.
Express a in terms of u

1.65 — Question 11. In perfect conditions, it takes 38 metres to stop a car driving at a speed of 80 kmh^{-1}.

a) Find the time it takes to stop assuming the deceleration is constant.

b) Comment on why this may not be an accurate answer for modelling the stopping of a car in a real-world scenario.

1.65 — Question 12. Ceri cycles 2 km on a straight horizontal section of a race. She is cycling 28 kmh^{-1} initially and finishes the section at 32 kmh^{-1}.

Find her acceleration for this section, assuming she accelerates at a constant rate.

1.65 — Question 13. A car travels along a straight horizontal road with constant acceleration, passing three points A, B and C. It takes 15 s to travel from B to C. It is moving at 12 ms^{-1} and 16 ms^{-1} at B and C respectively.

a) Find the acceleration of the car

b) Find the speed of the car at A given the distance between A and B is 150 metres.

1.65 — Question 14. A car is travelling along a straight horizontal road with a constant acceleration, passing three points A, B and C.
It takes 6 seconds to travel from A to B and a further 4 seconds to travel 10.4 metres from B to C.
It is travelling at 5 ms^{-1} as it passes point B. Find

a) the acceleration of the car.

b) the speed of the car at A.

c) how far the car travels beyond point C before it comes to rest.

1.65 — Question 15. A vehicle travels with constant acceleration along a straight flat road.
It passes three checkpoints A, B and C and in each case is travelling at 15 ms^{-1}, 30 ms^{-1} and 50 ms^{-1}.
Find

a) the ratio of the times taken to drive the distances AB and BC.
b) the distance between AB given the distance AC is 150 metres.

1.65 — Question 16. A truck is travelling along a straight horizontal road with constant acceleration.
A set of traffic lights is 25 metres away from the truck. A junction is a further 110 metres on from the traffic lights. The truck takes 2 seconds to travel to the traffic lights and 5 seconds to travel between the traffic lights and the junction.
Find

a) the acceleration of the truck.
b) the speed of the truck in its initial position.

1.65 — Question 17. Two cars, A and B, are travelling along a flat, straight road.
Car A is driving at u ms^{-1} throughout its motion.
Initially, car B is 15 metres behind car A and is driving 0.2 ms^{-1} slower than car A but is accelerating at 0.8 ms^{-2}.
Find the time it takes for Car B to catch up Car A.

1.65 — Question 18. A car is travelling on a straight flat road.
Initially it has velocity of 6 ms^{-1} and drives with constant acceleration, a ms^{-2}, past two checkpoints.
It reaches the first checkpoint which is at 48 metres from the initial point, in k seconds.

a) Find a giving your answer in terms of k
b) Find the initial speed for the second phase of the motion in terms of k

The next checkpoint is a further 204 metres away and it reaches it in 10 seconds.

c) Find the value of k
d) Find the value of a

1.65 — Question 19. A stone is dropped on a planet from a height of 3 metres. It takes 2.2 seconds to hit the surface. Find the acceleration due to gravity on this planet.

1.65 — Question 20. The following questions use $g = 9.81$ $\mathbf{ms}^{-2}$ A ball is dropped, and it takes 0.8 seconds to reach the ground. Find

a) the height above the ground from which the ball is dropped.
b) the speed in terms of t
c) the speed when the ball hits the ground.

1.65 — Question 21. The following questions use $g = 9.8$ **ms**$^{-2}$ An object, which is initially at rest is dropped from a height of 8m above the ground.

a) Calculate the speed of the object when it hits the ground.
b) Calculate the time it takes to hit the ground.
c) State a refinement to the model, apart from air resistance, which would make the model more realistic.

1.65 — Question 22. A ball is projected upwards from the ground with speed 14.7 ms^{-1}. Find

a) the time it takes to reach ground again.
b) the total distance travelled in the first 2 seconds of motion.

1.65 — Question 23. A pebble is thrown upwards in the air from the edge of a bridge with a velocity of 24.5 ms^{-1}.

a) Find the time it takes for the ball to reach its maximum height.

After 6.5 seconds from being thrown, the pebble lands in the river below. Find

b) the height of the bridge at the point where the pebble was thrown.

c) the total distance travelled by the pebble.

1.65 — Question 24. A stone is thrown vertically upwards in the air starting 1.65 m above the ground being projected with initial speed 6.25 ms^{-1}. The stone moves freely under gravity.

a) Find the time at which the stone hits the ground.

b) Find the times at which the stone is precisely 2 m above the ground.

c) Hence find the total time the stone spends more than 2 metres above the ground.

1.65 — Question 25. The following questions use $g = 10$ **ms**$^{-2}$ A ball is thrown vertically upwards with initial speed u ms^{-1} from a height of 15m above the ground. It hits the ground with speed 22 ms^{-1}

a) Find the initial speed.
b) What is the greatest height it reaches?
c) Find the time it takes to hit the ground.
d) Given the ball should be subject to air resistance, explain the how this would affect your answers to the previous parts of the question.

1.65 — Question 26. A parachutist steps off a plane at a height of 2800 m. They free-fall for 4 seconds before opening their parachute.

a) Find the speed of the parachutist the moment the parachute opens.

It is given that from the point the parachute opens, they decelerate at 12 ms^{-2} for 3 seconds before reaching a constant speed.

b) Find the total time it takes to reach the ground from initially stepping off the plane.

1.65 — Question 27. A ball is thrown vertically upwards from the ground at 20 ms^{-1} whilst simultaneously a second identical ball is thrown vertically downwards at 5 ms^{-1} from a height of 10 metres above the ground. Find

a) the time at which both balls are at the same height.
b) the distance from the ground at which both balls are at the same height.

1.65 — Question 28. A ball A is thrown vertically upwards from the ground at 20 ms^{-1} whilst simultaneously a second identical ball B is thrown vertically upwards at 12 ms^{-1} from a height of 5 metres above the ground. Find

a) the amount of time that both balls are above the height of the initial start point of ball B.
b) the time at which both balls are at the same height.
c) the distance from the ground at which both balls are at the same height.

1.65 — Question 29. The following questions use g in exact form An object is dropped from a height of h metres. Show that $v = \sqrt{2gh}$

1.65 — Question 30. A stone is dropped from a height of h metres above horizontal ground. It falls freely downwards. Its speed when reaching the ground is 8 ms^{-1}. Show that $h = \frac{32}{g}$

1.65 — Question 31. A ball is projected directly upwards at u ms^{-1}. Find an expression for

a) the maximum height in terms of u

b) the time it takes to return to its start point in terms of u

1.65 — Question 32. Two particles A and B are released from rest. A is released from k metres above a horizontal surface. B is released 0.5 seconds later from $\frac{k}{10}$ metres above the horizontal surface. They land simultaneously on the surface at t seconds after the release of A.

Find the value of t and show that $k \approx 0.267g$

1.71 — Question 1. A particle moves along a straight line so that its displacement s metres at time t seconds, where $t \geq 0$, is given by each of the following functions. Find expressions for the velocity and the acceleration of the object at time t for:

a) $s = 4t^3 - 5t$
b) $s = t^5 + 2t^2$
c) $s = t(t-1)$
d) $s = 2t - \frac{1}{t}$

1.71 — Question 2. A particle starts at rest before moving along a straight line. Its velocity v ms^{-1} at time t seconds is given by each of the following functions, where $t \geq 0$. In each case find the displacement in terms of t:

a) $v = 8t^3 - 6t$
b) $v = 6t^2 + 2t^3$
c) $v = \frac{1}{10}t^2(t-5)$
d) $v = t - \frac{1}{t^2}$

1.71 — Question 3. A particle is moving along a straight line. Its acceleration a ms^{-2} at time t seconds is given by each of the following functions, where $t \geq 0$.
In each case find the velocity at t in terms of t, given that is known that the particle is moving with velocity v_1 at time t_1

a) $a = 15t^4 - 4t, t_1 = 1, v_1 = 2$
b) $a = 9t^2 + 5t^4, t_1 = 2, v_1 = 40$
c) $a = -0.09t(t - 30), t_1 = 1, v_1 = 3$
d) $a = \sqrt{t} - \frac{1}{t^3}, t_1 = 1, v_1 = 2$

1.71 — Question 4. A car drives between two sets of traffic lights, stopping at both. Its speed v ms^{-1} at time t s is modelled by

$$v = -\frac{1}{50}t^2 + \frac{6}{5}t, \qquad 0 \leq t \leq 60$$

a) Find the times at which the car is stationary.
b) Sketch the velocity-time graph for the motion of the car.
c) Find the distance between the two sets of traffic lights.
d) Find an expression for acceleration a ms^{-2} in terms of t
e) Explain why the model is no longer appropriate after 60 seconds.

1.71 — Question 5. A particle's acceleration a ms^{-2} with time t seconds is given by

$$a = 15\sqrt{t} + 2, \qquad t \geq 0$$

The initial velocity is 1 ms^{-1} and when $t = 1$ the displacement of the particle is 6 m.

a) Find an expression for the velocity v with time t

b) Find an expression for the displacement s with time t

1.71 — Question 6. A particle's acceleration a ms^{-1} at any given time t seconds is given by

$$a = -30t^2 + 25t^3, \qquad t \geq 0$$

The particle initially starts at position O, with a velocity of 5 ms^{-1}.

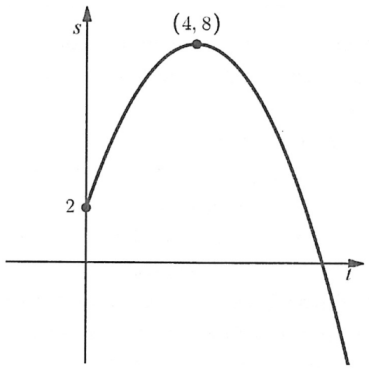
a) Find an expression for the velocity given the particle initially starts at position O.

b) Find the displacement of the particle when 2 seconds have passed.

1.71 — Question 7. The displacement s (metres) over time t (seconds) for a particle is given by

$$s = -\frac{3}{8}t^2 + kt + 2, \qquad t \geq 0$$

where k is a positive constant and is shown in the diagram below.



a) Show that $k = 3$
b) State the distance travelled by the particle when $t = 4$
c) Find the time at which the particle returns to its initial position.
d) Find an expression for the velocity v in terms of t
e) Find the speed of the particle when $t = 10$ seconds.

1.71 — Question 8. Prerequisite knowledge for Q8: $F_{\text{net}} = ma$ An object of mass 0.4 kg is moving in a straight line under the action of a resultant force F N. Its velocity v ms^{-1} at any given time t seconds may be given by

$$v = 8t - 3t^2, \qquad t \geq 0$$

a) Find an expression for the object's displacement, s metres, in terms of t
b) Find an expression for the object's acceleration, a ms^{-2}, in terms of t
c) Find an expression for F in terms of t

1.71 — Question 9. The velocity, v ms^{-1} at any given time t seconds of a car may be given by

$$v = kt - 3 + \frac{1}{t^2}, \qquad \frac{1}{2} \leq t \leq 10$$

where k is an unknown positive constant.
It is known that the acceleration when $t = 2$ is 1.75 ms^{-2}.

a) Find the value of k
b) Find the total distance travelled by the car for $\frac{1}{2} \leq t \leq 10$

1.71 — Question 10. Ellis is going to the bus stop. His displacement, s metres at any given time t seconds is

$$s = -\frac{1}{k}t(t - 240), \qquad 0 < t \leq 120$$

where k is an unknown positive constant.
It is known that the velocity when $t = 20$ is 4 ms^{-1}.

a) Show that $k = 50$
b) Find the speed at which Ellis is travelling when $t = 60$ seconds
c) Find the speed at which Ellis is travelling when $t = 120$ seconds.
d) Explain in context what happens to Ellis' velocity and displacement if this same model is used for $t \geq 120$

1.71 — Question 11. A particle's velocity, v ms^{-1} with time t seconds is given by

$$v = -t + 10, \qquad t \geq 0$$

Find when the particle will return to its original position.

1.71 — Question 12. An object moves along a straight line so that its acceleration a ms^{-2} at time t seconds is given by

$$a = 4 + 3t - 6t^2, \qquad t \geq 0$$

a) Given that the object is initially stationary, find the velocity, v, in terms of t
b) Given that when 1 second has passed, the object has moved in a positive direction of 6 metres from its initial position, find the displacement, s in, in terms of t
c) Hence, find the time at which the object returns to its original position, giving your answer to 3 significant figures.

1.71 — Question 13. A particle's displacement, s metres with time t seconds is given by

$$s = t^3 - 6t^2 + 12t, \qquad 0 \leq t \leq 10$$

a) Find the distance travelled by the particle over the 10 seconds.
b) Find the time when the particle is at rest.
c) Show that, according to this model, the particle will never reverse direction.
d) Show that it does not have a constant acceleration.

1.71 — Question 14. A particle's acceleration, a ms^{-2} with time t seconds is given by

$$a = -1.8t + 18$$

where $t \geq 0$
Find the time it takes for the particle to travel 600 metres given it starts from rest.

1.71 — Question 15. A car is being driven on a straight flat road, with velocity v ms^{-1} at time t seconds is given by

$$v = 2t - 0.001t^3, \qquad 0 \leq t \leq 25$$

a) Find the distance travelled by the car in the first 20 seconds.
b) Show that for $0 \leq t \leq 25$, the car is always accelerating (and not decelerating).

1.71 — Question 16. The displacement, s metres at any given time t seconds of a car may be given by

$$s = \frac{4}{15}t^2 - \frac{1}{675}t^3, \qquad 0 \leq t \leq 120$$

a) Find the velocity of the car v in terms of t
b) Find the maximum speed for $0 \leq t \leq 120$

1.71 — Question 17. Prerequisite knowledge for Q17-Q18: Constant Acceleration A car is being tested on a straight, flat road. Its velocity is found to be 12.5 ms^{-1} at 5 seconds and 25 ms^{-1} at 10 seconds. A model for its velocity v ms^{-1} against time t seconds is

$$v = at^2 + bt^3, \qquad t \geq 0$$

where a and b are unknown constants.

a) Find the values of a and b

A second model suggests that the cars motion may be modelled with constant acceleration.

b) Show that the distance travelled for both these two models in the first 10 seconds is the same.

1.71 — Question 18. A runner having started from rest reaches a velocity of 3.5 ms^{-1} in 7 seconds.

a) Assuming constant acceleration, find a model for the velocity v ms^{-1} at any given time t

b) Find the distance travelled by the runner in the 7 second duration using this model.

An alternative model is given by

$$v = -\frac{1}{k}(t-7)^2 + 3.5, \qquad t \geq 0$$

where k is a positive constant.

c) Find the value of k

d) Find the distance travelled by the runner in the 7 second duration according to this second model.

e) By first sketching both models on a graph, explain why the second model gives a larger distance than the first.

1.71 — Question 19. The velocity, v ms^{-1} at any given time t seconds of a cyclist for a 6 second section of their journey may be given by

$$v = \frac{2}{5}t^3 - 4t^2 + 10t, \qquad 0 \leq t \leq 6$$

a) Find the distance the cyclist has travelled in the first 5 seconds.
b) Find the maximum speed of the cyclist and the time at which it occurs.
c) Find the minimum acceleration of the cyclist and state what its value represents in context of the cyclist.
d) Explain why this model would not be valid in the long term.

1.71 — Question 20. The velocity, v ms^{-1} at any given time t seconds of a person walking along a flat, straight path may be given by

$$v = \frac{k}{\sqrt{t}} + \frac{t}{80}, \qquad 0 < t \leq 120$$

where k is an unknown positive constant.
It is known that the acceleration when $t = 25$ is 0.0065 ms^{-2}.

a) Show that $k = \frac{3}{2}$
b) Find the total distance travelled by the person for $60 < t \leq 120$
c) Comment on the validity of the model for small values of t

1.71 — Question 21. The velocity, v ms^{-1} at any given time t seconds of a particle may be given by

$$v = pt^2 + qt + r, \qquad t \geq 0$$

where p, q and r are unknown constants.
Initially, it has a velocity $v = 7.8$ ms^{-1}. It reaches its minimum velocity of 3 ms^{-1} when $t = 4$
Find the values of p, q and r

1.71 — Question 22. The acceleration, a ms^{-2} at any given time t seconds of a particle may be given by

$$a = kt - 4, \qquad t \geq 0$$

where k is an unknown constant. The initial velocity is twice the magnitude of the minimum velocity which occurs when $t = 2$
Find the value of k and hence an expression for the velocity, v in terms of t only.

1.71 — Question 23. The velocity, v ms^{-1} at any given time, t seconds, of a particle may be given by

$$v = \frac{3}{2}t^2 - \frac{9}{2}t + 3, \qquad t \geq 0$$

The particle is initially at the origin.
Determine the displacement of the particle from the origin when it has covered a total distance of 8.5 metres.

1.71 — Question 24. The velocity v ms^{-1} at any given time t seconds of a particle starting at initial position O may be given by

$$v = -t^2 + kt, \qquad 0 \leq t \leq 9$$

where k is an unknown positive constant.
Initially, the acceleration is 6 ms^{-2}.

a) Show that $k = 6$
b) Show that the maximum velocity occurs when $t = 3$
c) Find the distance it has travelled when it has reached its maximum velocity.
d) Find the displacement of the particle over the 9 second duration.
e) Find the time at which the particle is not moving.
f) Find the distance travelled over the 9 second duration.

1.71 — Question 25. The velocity v ms^{-1} at any given time t seconds of a cyclist may be given by

$$v = \begin{cases} 0.81t^2 - 0.09t^3 & 0 \leq t \leq 6 \\ 9.72 & 6 \leq t \leq 20 \end{cases}$$

a) State the initial velocity.
b) Describe the motion of the cyclist.
c) Find the maximum acceleration of the cyclist.

1.71 — Question 26. Filip is walking to a shop which is 800 m from his house, but on arriving he realises he left his phone at home and returns to retrieve it. His displacement from his house s m at any given time t seconds may be given by

$$s = \begin{cases} \frac{3}{200}t^2 - \frac{1}{40000}t^3 & 0 \leq t \leq t_1 \\ -3.2t + 2080 & t_1 \leq t \leq t_2 \end{cases}$$

where t_1 is the time at which Filip arrives at the shop, and t_2 is the time at which Filip arrives home. According to the model,

a) find the time t_1 it takes to reach the shop.
b) find the time t_2 taken for the whole journey, giving your answer to the nearest second.
c) find the maximum velocity during his journey to the shop.
d) find the maximum speed for his full journey.
e) Comment on the validity of the model for the return journey.

1.71 — Question 27. Prerequisite knowledge in Q27-Q28: Differentiation of the form ae^{kx} A school is holding a 20 m race. The displacement s m at any given time t seconds of the fastest child running the race is given by

$$s = 6t - 15 + 15e^{-0.4t}, \qquad 0 \leq t \leq k$$

a) Show that the time k seconds according to the model, at which the fastest child finishes the 20 m is approximately 5.56 s.
b) Find an expression for the velocity v ms^{-1} in terms of t
c) Find the initial velocity of the child.
d) Find the maximum acceleration a ms^{-2} of the child during the race.

1.71 — Question 28. The vertical displacement s metres above ground level of a carriage in "The Drop", a ride at an amusement park, at any given time t in the first 40 seconds may be given by

$$s = 0.8e^{0.1t} + 8e^{-0.2t}, \qquad 0 \leq t \leq 40$$

The ride stops at 40 seconds before dropping.
a) State the height above the ground at which the ride begins.
b) Find the time at which the ride changes direction, giving your answer to the nearest second.
c) Find the distance the ride drops before travelling upwards again.
d) Find an expression for the vertical acceleration a ms^{-2} in terms of t.
e) Find the maximum vertical speed of the ride according to the model.

In the second phase of the ride at $t = 40$, the ride drops in free-fall for 2 seconds before applying a brake.
f) Using $g = 9.8$ ms^{-2} determine the distance the carriage drops before the brake is applied.

1.71 — Question 29. Prerequisite knowledge for Q29: $F_{\text{net}} = ma$ An object of mass 0.2 kg is moving in a straight line under the action of a resultant force F N. Its displacement s m at any given time t seconds may be given by

$$s = -\frac{1}{10}t^2(t-4), \qquad t \geq 0$$

Find an expression for F in terms of t